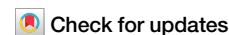


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Manipulating the crystallization kinetics of halide perovskites for large-area solar modules



Zhaojin Wang^{1,2}, Xiao Duan^{1,2}, Jing Zhang³, Wenbin Yuan^{1,2}, Dinghao Qu^{1,2}, You Chen^{1,2}, Lijuan He^{1,2}, Haoran Wang^{1,2}, Guang Yang⁴, Wei Zhang⁵✉, Yang Bai^{1,2,5}✉ & Hui-Ming Cheng^{1,2,5}✉

In the last decade, laboratory-scale single-junction perovskite solar cells have achieved a remarkable power conversion efficiency exceeding 26.1%. However, the transition to industrial-scale production has unveiled a significant efficiency gap. The central challenge lies in the difficulty of achieving uniform, high-quality perovskite films on a large scale. To tackle this issue, various innovative strategies for manipulating crystallization have emerged in recent years. Based on an in-depth fundamental understanding of the nucleation and growth mechanisms in large-area perovskite films prepared through blade/slot-die coating methods, this review offers a critical examination of crystallization manipulation strategies for large-area perovskite solar modules. Lastly, we explore future avenues aimed at enhancing the efficiency and stability of large-area PSMs, thereby steering the field toward commercially viable applications.

Halide perovskites have garnered considerable attention to their high absorption coefficient, long carrier diffusion length, solution processability, and cost-effectiveness^{1–5}. These attributes have propelled the power conversion efficiency (PCE) of laboratory-scale perovskite solar cells (PSCs) from 3.8 to 26.1% over the past decade^{6–10} (<https://www.nrel.gov/pv/interactive-cell-efficiency.html>), a figure comparable to that of single-crystalline silicon solar cells. This progress underscores their significant potential for practical applications (<https://www.nrel.gov/pv/interactive-cell-efficiency.html>)¹¹, owing

Despite the rapid advancement in lab-scale PSCs, the PCE of large-area perovskite solar modules (PSMs) lags behind. This disparity can be attributed to the processing method, more defects, interconnection between sub-modules, high sheet resistance, etc. (has been well-reviewed)^{12–16}. Among them, the most important loss comes from the processing method. This is because of the fabrication of high-performance lab-scale PSCs using spin-coating^{6–10}, a technique unsuitable for PSM production due to non-uniform film formation resulting from inconsistent centrifugal and centripetal forces^{13–16}. To overcome this challenge, alternative large-area PSM fabrication methods, such as inkjet printing, screen printing, spray coating, blade-coating, and slot-die coating, have been explored^{13–16}. Among these, blade-coating and slot-die coating have emerged as promising solutions due to their

simplicity, high material utilization rate, low cost, and rapid production rate^{13–16}. The PCE and stability development of PSMs by blade/slot-die coating methods in recent years is shown in Fig. 1a, b (Generally, PSMs are categorized based on their size (<https://www.nrel.gov/pv/interactive-cell-efficiency.html>)¹²: lab-scale cells are smaller than 10 cm², mini-module from 10 to 200 cm², sub-module from 200 to 800 cm², small-module from 800 to 6500 cm², standard-module from 6500 to 14,000 cm² and large module exceed 14,000 cm²) (<https://www.gcl-power.com/>; <https://www.microquanta.com/>; <https://mellow-energy.cn/official/journalism/details/24/24/109.html>; <https://www.utmilight.com/>; https://www.renshinesolar.com/page99?article_id=112; <https://www.oxfordpv.com/news/oxford-pv-sets-new-solar-cell-world-record>; <https://www.oxfordpv.com/news/oxford-pv-sets-new-solar-panel-efficiency-world-record-0>)^{17–31}.

While large-area PSMs fabricated through blade/slot-die coating methods have made significant progress, their efficiency still lags behind commercial crystalline silicon (c-Si) modules. For instance, the PCE of a large perovskite module reaches 19.04% at a designated area of 20,000 cm², while a c-Si large module achieves 24.7% at a designated area of 17,806 cm² (see ref. 12 and <https://www.gcl-power.com/>). Therefore, to facilitate its industrialization in the near future, there is an urgent need to further improve the efficiency of large-area PSMs.

¹Faculty of Materials Science and Energy Engineering, Shenzhen University of Advanced Technology, Shenzhen, China. ²Institute of Technology for Carbon Neutrality, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China. ³Advanced Technology Institute (ATI), University of Surrey, Guildford, Surrey, UK. ⁴Department of Electrical and Electronic Engineering, Photonic Research Institute (PRI), Research Institute of Smart Energy (RISE), Research Institute for Advanced Manufacturing (RIAM), The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong, China. ⁵Shenzhen Key Laboratory of Energy Materials for Carbon Neutrality, Shenzhen, China. ✉e-mail: wz0003@surrey.ac.uk; y.bai@siat.ac.cn; hm.cheng@siat.ac.cn

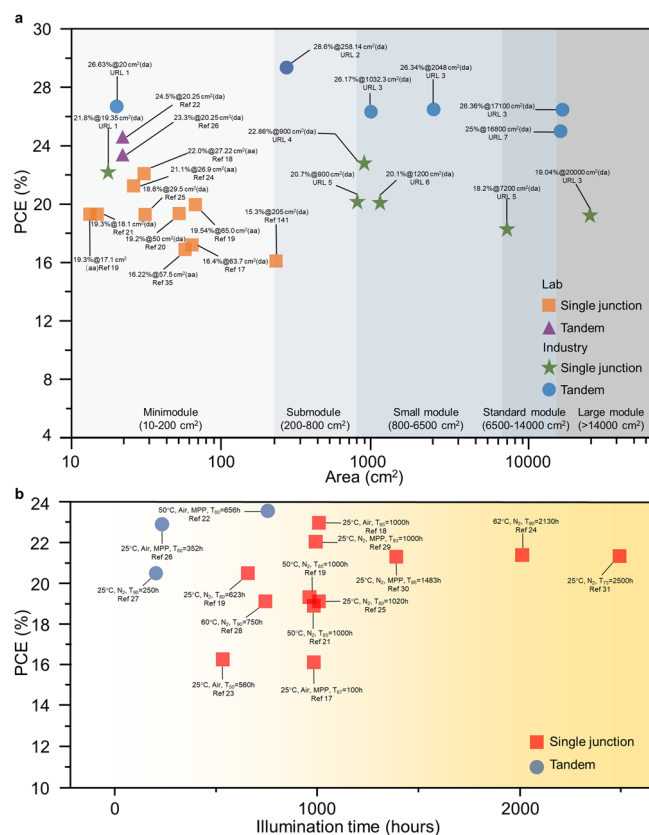


Fig. 1 | The PCE development of PSMs by blade/slot-die coating method. a PCE progress. Due to factors such as experimental equipment, environment, raw material purity, and so on, the PCE of the academic community lags behind that of the industry. aa is active area, da is designated area. **b** Stability progress. Due to the problems of phase separation and current density mismatch, the stability of the tandem module is significantly lower than that of the single-junction module. <https://www.microquanta.com/>. <https://www.oxfordpv.com/news/oxford-pv-sets-new-solar-cell-world-record>. <https://www.gcl-power.com/>. <https://mellow-energy.cn/official/journalism/details/24/24/109.html>. <https://www.utmilight.com/>. https://www.renshinesolar.com/page99?article_id=112. <https://www.oxfordpv.com/news/oxford-pv-sets-new-solar-panel-efficiency-world-record-0>.

Numerous factors influence the efficiency of PSMs, including perovskite film quality, sub-cell width, dead zone, optical loss, module structure, interface defects, and encapsulation^{2,12–14,16}. Among these, the crystal quality and uniformity of large-area perovskite films have been recognized as the most critical factor, primarily dictated by crystallization kinetics^{2,12–14,16}. A plethora of studies have been conducted to accelerate nucleation through interface engineering and solvent engineering, and significant advancements have been achieved in the past few years^{17,32–36}. In addition, great progress has also been made in delaying growth via additive engineering^{18,19,37}. For instance, Ding et al. regulated the crystallization kinetics by additive engineering (methylammonium chloride and 1,3-bis(-cyanomethyl) imidazolium chloride) to achieve fast nucleation and slow growth, obtained a mini-module with an aperture area of 27.22 cm² and a certified PCE of 23.30%¹⁸. Hence, there is an urgent need to systematically investigate and elucidate the crystallization kinetics of large-area PSMs to further enhance their efficiency.

Previous reviews on crystallization kinetics primarily concentrated on lab-scale PSCs^{38–46}. However, lab-scale PSCs predominantly utilize the spin-coating method, whereas large-area PSMs are normally fabricated using the blade/slot-die coating method. These distinct fabrication methods result in notable differences in the crystallization process, rendering the crystallization kinetics manipulation strategies tailored for lab-scale PSCs inadequate for large-area PSMs. Despite earlier review articles on large-area PSMs

covering aspects such as cost analysis, device structure, solvent engineering, and ink engineering^{2,12–14,16,20,47–54}, there remains a gap in providing a systematic and timely overview specifically addressing the manipulation of nucleation and growth in large-area perovskite film fabrication, which is crucial for further enhancing the performance of PSMs.

To bridge this gap, this review meticulously examines the crystallization kinetics of large-area perovskite films fabricated using blade/slot-die coating methods. We first elucidate the parameters that impact the nucleation and growth process, including temperature, concentration, interface energy, and solute precipitation rate. Subsequently, we provide a critical overview of recent advances in modulating the crystallization kinetics of large-area perovskite films fabricated employing blade/slot-die coating techniques. Finally, we give a brief outlook discussing the remaining challenges facing perovskite crystallization manipulation and outlining future directions for closing the gap between lab-scale PSCs and fab-scale PSMs.

Crystal nucleation and growth

Nucleation

In the classical crystal growth theory, the formation of crystals generally involves two processes: nucleation and growth. In the classical nucleation theory, it is usually assumed that the formed nuclei are spherical⁵⁵. In preparing large-area PSMs by blade/slot-die coating methods, crystal nucleation usually belongs to heterogeneous nucleation^{56,57}. Therefore, according to Young's equation and Thomson–Gibbs equation⁵⁵, the specific expression of heterogeneous nucleation energy barrier can be expressed as follows^{55–57}:

$$\Delta G_{\text{hetero}}^* = \frac{16\pi\sigma^3v^2}{3\Delta\mu^2} \frac{2 - 3\cos\theta + \cos^3\theta}{4} \quad (1)$$

where $\Delta G_{\text{hetero}}^*$ is the energy barrier of heterogeneous nucleation, v is the critical nucleus volume, θ is the contact angle between solution and substrate, $\Delta\mu$ is the chemical potential difference between the precipitated crystal and the mother liquid, and σ is the interface energy between the solution and the substrate.

The expression of the nucleation rate can be obtained as follows^{55–57}:

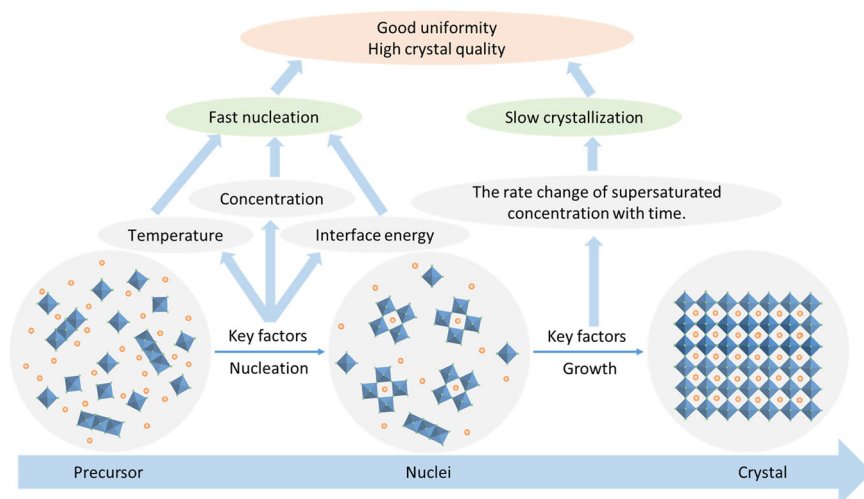
$$\frac{dN_{\text{hetero}}}{dt} = \Gamma \exp\left[\frac{-\Delta G_{\text{hetero}}}{k_B T}\right] = \Gamma \exp\left[\frac{-16\pi\sigma^3v^2}{3k_B T\Delta\mu^2} \frac{2 - 3\cos\theta + \cos^3\theta}{4}\right] \quad (2)$$

where N_{hetero}^* is the heterogeneous nucleation rate, t is the time, T is the temperature, k_B is the Boltzmann constant, and Γ is the Zeldovich factor. From the nucleation energy barrier and nucleation rate equations, the main factors affecting the nucleation rate are concentration, temperature, and interface energy. Increasing the nuclei and nucleation rate is necessary to achieve a large-area perovskite film with good compactness and uniformity. According to Eqs. (1) and (2), the heterogeneous nucleation rate and the number of nuclei can be increased by increasing the temperature, the solution's concentration, and the interface energy. Although increasing the nucleation rate and the number of nuclei can improve the compactness and uniformity of large-area perovskite films, it will also accelerate crystal growth and is not conducive to improving crystal quality. Therefore, we will discuss how to delay growth to improve the crystal quality.

Growth

The compactness and uniformity of large-area perovskite films depend on the nucleation rate and the number of nuclei, while the crystal quality depends on the growth rate. Once the nucleus is formed, the growth begins spontaneously. Here, we assume that the size of the formed crystal nucleus is uniform. According to McCabe's law, the total crystal growth rate can be

Fig. 2 | Nucleation and growth diagram of large-area perovskite thin films prepared by blade/slot-die coating methods. The key factors affecting the nucleation are temperature, concentration and interface energy, and the key factor affecting the growth is the rate change of the solution supersaturation concentration with time. Fast nucleation and slow crystallization are essential to improve large-area perovskite films' compactness, uniformity, and crystal quality.



expressed as follows^{55–57}:

$$R = -\frac{d\Delta C}{dt} \quad (3)$$

Where R is the total crystal growth rate, t is the time, and ΔC is the supersaturation concentration of the solution. According to Eq. (3), delaying the solute precipitation rate can reduce the rate change of the solution supersaturation concentration with time, thereby slowing crystal growth. It is worth mentioning that the number of nuclei will also affect the crystallization rate. Although many crystal nuclei are beneficial for improving the uniformity of large-area perovskite films, they will also lead to too fast crystal growth and reduce crystal quality. Therefore, delaying crystallization is crucial for improving the crystal quality of large-area perovskite films.

In summary, increasing the nucleation rate and the number of nuclei in the nucleation stage is beneficial to improve the uniformity and compactness of the film, and delaying the growth in the growth stage is beneficial to improve the crystal quality to achieve the preparation of large-area perovskite films with good compactness, good uniformity, and high crystal quality. Next, we discuss how to achieve fast nucleation and slow growth (Fig. 2).

Fast nucleation and slow growth

Fast nucleation

The key factors affecting the nucleation of large-area perovskite films mentioned above are temperature, concentration, and interface energy. Therefore, we will discuss how to regulate the nucleation process through temperature engineering, concentration engineering, and interface energy engineering, respectively.

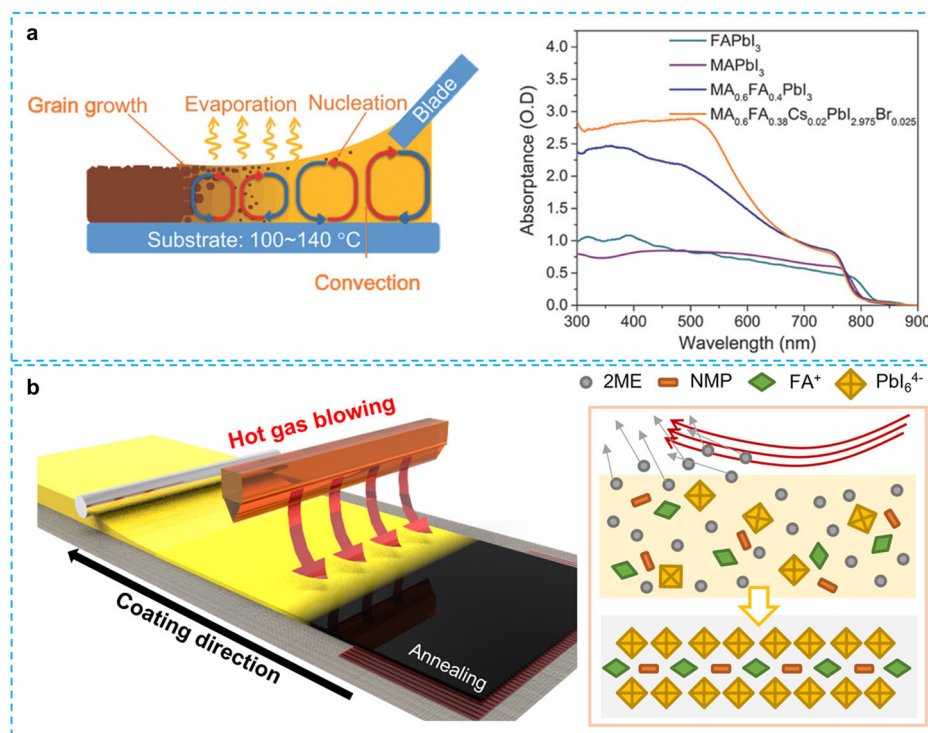
Temperature engineering. The commonly used strategy in temperature engineering is to increase the substrate temperature. Increasing the substrate temperature has two effects. As aforementioned, elevated temperatures reduce the nucleation energy barrier and increase the nucleation rate, resulting in more crystal nuclei. Higher substrate temperatures also hasten solvent evaporation, quickening nucleation and enhancing large-area perovskite film compactness and uniformity. For example, Tang et al. found that increasing the substrate temperature is beneficial to improve the density and uniformity of the film when preparing a variety of perovskite films by the blade-coating method (Fig. 3a)⁵⁸. Ren et al. reported that a mini-module of a carbon electrode with an active area of 50 cm² and a PCE of 17.05% was prepared by blade-coating at a substrate temperature of 120 °C⁵⁹. Moreover,

the optimization of the N₂ knife temperature in the blade-coating method has been shown to be beneficial, as evidenced by Han et al.'s technique, which utilizes a heated N₂ knife to aid in the preparation of PSMs, resulting in a sub-module with an active area of 400 cm² and a PCE of 15.43% (Fig. 3b)⁶⁰. However, the approach of temperature engineering is not without its drawbacks. While a higher substrate/knife temperature may expedite nucleation and the formation of numerous nuclei, it also precipitates rapid growth during the coating stage rather than during annealing. Consequently, nucleation and growth occur simultaneously during the coating process, potentially leading to an excessively rapid crystal growth rate that is detrimental to the film's crystal quality.

Concentration engineering. Concentration engineering mainly discusses how to quickly supersaturate the solution concentration in the crystal nucleation stage, thereby accelerating nucleation and forming many crystal nuclei. The main methods are the antisolvent method (AS), gas-assisted solution process method (GASP), vacuum flash-assisted solution process method (VASP), and solvent engineering. Next, we discuss these four methods respectively.

AS. The antisolvent (AS) method, a cornerstone in the spin-coating fabrication of lab-scale perovskite solar cells^{61–73}, employs a nonpolar solvent such as chlorobenzene to interact with the typically polar perovskite precursor solvent, like N, N-dimethylformamide (DMF)^{72,73}. This interaction is governed by the principle of similar miscibility, which posits that solvents with different polarities mix poorly¹⁵. Consequently, the introduction of an antisolvent during spin-coating precipitates instantaneous supersaturation of the precursor wet film, leading to rapid nucleation. Expanding this principle to large-area films, the AS method similarly induces quick supersaturation in perovskite films prepared via blade/slot-die coating. This rapid supersaturation is instrumental in accelerating nucleation, which in turn enhances the film's compactness and uniformity. Illustrating this, Yang et al. demonstrated that a mini-module with an active area of 20.77 cm², boasting a certified quasi-stabilized PCE of 16.63%, was successfully fabricated using the slot-die coating coupled with AS method⁷⁴. It should be emphasized that the rapid formation of a large number of nuclei by antisolvent can improve the uniformity of large-area films, but it also leads to an excessive growth rate. Therefore, they added diphenyl sulfoxide (DPSO) to delay crystallization and improve crystal quality⁷⁴. Despite these advancements, the AS method has limitations. The substantial volumes of antisolvent required can escalate both the costs associated with the commercialization of PSMs and their environmental footprint, posing challenges that warrant careful consideration.

Fig. 3 | Fast nucleation by temperature engineering. **a** The GASP combined with blade-coating to prepare a mini-module. Reprinted with permission from ref. 58, copyright 2017 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. **b** is the experimental process diagram and photograph of the mini-module prepared by the blade-coating method combined with VASP. Reprinted with permission from ref. 60, copyright 2023 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.



GASP. The principle of GASP is that the flowing atmosphere can accelerate the volatilization of the solvent, make the precursor wet film supersaturated quickly, accelerate the nucleation, and increase the number of nuclei, thereby improving the compactness and uniformity of the large-area perovskite film. For example, Deng et al. reported that a mini-module with an aperture area of 63.7 cm² and a certified module efficiency of 16.4% was prepared by N₂-assisted blade-coating (Fig. 4a)³². Although GASP can promote rapid nucleation and improve uniformity, to cope with the problem of excessive crystallization caused by a large number of nuclei, they introduced dimethyl sulfoxide (DMSO), formamidinium chloride (FACl), phenylethylammonium chloride (PEACl), and formamidinium hypophosphite (FAH₂PO₂) into the solution to delay crystallization³². The GASP method has been widely used in blade/slot-die coating, which is conducive to rapid coating at room temperature and is suitable for low-cost preparation of large-area perovskite films^{17,32–35}. However, the GASP method has obvious limitations, such as easily leading to a ribbing effect, and reducing the uniformity of the film. In addition, the GASP method requires strict control of blowing conditions, such as gas pressure, nozzle angle and airflow.

VASP. The VASP emerges as a pivotal technology for perovskite film fabrication, wherein the wet precursor film is subjected to a rapid high vacuum within a vacuum chamber. This process, typically completed within one minute, effectively removes the solvent, thereby initiating swift nucleation and the formation of numerous nuclei. As a result, VASP facilitates the creation of dense and homogeneous large-area perovskite films. Traditional antisolvent & spin-coating methods grapple with the challenge of fully extracting DMSO due to its high boiling point^{21,75–85}. The residual DMSO near the bottom interface leads to the formation of pinholes, detrimentally impacting the solar cells' efficiency and stability^{21,75–85}. While interface and additive engineering strategies have addressed this issue to some extent, they introduce complexity, particularly in large-area films^{21,75–85}. VASP offers a promising solution by ensuring the complete removal of DMSO, thereby obviating the need for additional engineering at the interface^{66–70}. This method not only resolves the issue of pinholes at the bottom interface but also simplifies the overall process, as shown in Fig. 4b, enhancing both the performance and the manufacturability of PSCs^{86–90}. For example, Hu et al.

reported the preparation of FAPbI₃ PSMs by VASP combined with the blade-coating method⁸⁷. Without introducing additives, the residual DMSO and pinholes at the bottom interface can be eliminated, and the compactness and uniformity of the film can be improved. Finally, a mini-module with an aperture area of 12.25 cm² and a PCE of 18.3% was obtained (Fig. 4b)⁸⁷. The VASP method has many advantages, such as fast and controllable removal of solvents to accelerate nucleation. After the VASP process, an intermediate phase film is usually formed. During the annealing process, this intermediate phase will improve the crystal quality and orientation of the film. The VASP can completely remove the strong Lewis bases (sLBAs) without coordination with perovskite, thereby reducing the pinholes and defects of the bottom interface. The disadvantage of the VASP method is that preparing large-area PSMs requires additional large-area vacuum equipment, which increases the equipment cost. One disadvantage of the VASP method is that preparing large-area PSMs requires an additional large-area vacuum equipment, which increases the equipment cost. In addition, the VASP method is also sensitive to the time window, which includes three time windows. The time window Δt₁ is from the end of blade/slot-die coating to the start of VASP, the time window Δt₂ is the vacuum time of VASP, and the time window Δt₃ is from the end of VASP to the start of annealing. The time windows Δt₂ and Δt₃ can be optimized and improved by empirical parameters. The time window Δt₁ for the same film may be a different time window (similar to the antisolvent spin-coating method), which may lead to non-uniformity in large-area perovskite films. To tackle this problem, recent efforts have been devoted to lengthening the time window Δt₁ by introducing additives. For example, 1,3-dimethyl-3,4,5,6-tetrahydro-2 (1H)-pyrimidinone (DMPU) was used instead of DMSO to delay the time window Δt₁⁹¹. Aminoacetamide hydrochloride (AAH) was also proven to effectively delay the time window Δt₁²². Despite these advances, further study is still needed to optimize the VASP technology, promoting the mass production of PSMs for practical application.

Solvent engineering. In the production of large-area perovskite thin films via blade/slot-die coating methods, the selection of solvents plays a critical role. Opting for highly volatile, weakly coordinating solvents, such as 2-methoxyethanol (2-ME), over moderately volatile, coordinating solvents

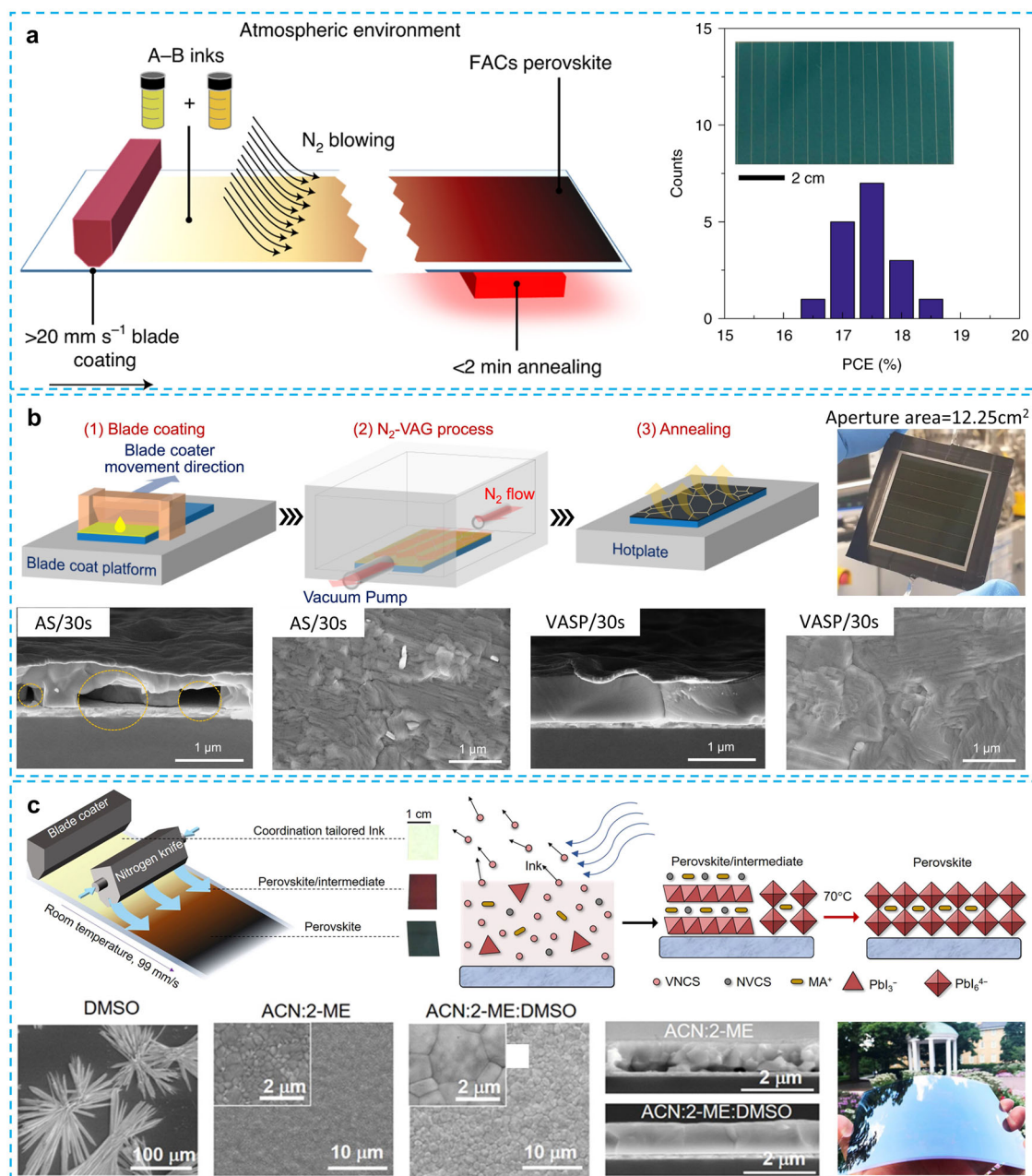


Fig. 4 | Fast nucleation by concentration engineering. **a** The GASP combined with blade-coating to prepare a mini-module. Reprinted with permission from ref. 32, copyright 2021 Springer Nature. **b** is the experimental process diagram and photograph of the mini-module prepared by the blade-coating method combined with VASP. Reprinted with permission from ref. 87, copyright 2023 Elsevier. **c** By using

ACN/2-ME to regulate the coordination and volatilization rate of the solvent for high-speed, room-temperature blading of perovskite photovoltaic films. Reprinted with permission from ref. 17, copyright 2019 American Association for the Advancement of Science.

like N, N-Dimethylformamide (DMF), can significantly accelerate solvent evaporation^{16,17,32–35}. This hastened volatilization leads to rapid supersaturation of the solution, thereby accelerating the nucleation rate and increasing the density of crystal nuclei. Consequently, a notable enhancement in the film's uniformity and compactness can be observed. However, the use of solely volatile weak coordinating solvents presents a dilemma: the solubility of perovskite precursors is compromised due to their weak coordination ability. To address this, a judicious amount of a less volatile, strongly coordinating solvent, such as DMSO, is often incorporated. This addition boosts the solubility of the raw materials and elevates the solution concentration, ensuring the formation of high-quality perovskite films. This nuanced balance between solvent volatility and coordination strength is pivotal in refining the film formation process, ultimately contributing to the

advancement of resultant perovskite devices. For example, Deng et al. added some volatile and weakly coordinated acetonitrile (ACN)/2-ME solvents to partially replace the DMSO solvent, which can accelerate the solvent evaporation rate to accelerate the nucleation (Fig. 4c)¹⁷. Although ACN/2-ME solvents can accelerate nucleation and improve the uniformity of large-area films, only using ACN/2-ME solvents will lead to crystal growth that is too fast, thus reducing the crystal quality of large-area films. Therefore, Deng et al. introduced a small amount of DMSO to delay crystal growth in ACN/2-ME solvents¹⁷. This ternary solvent strategy enables rapid preparation of large-area perovskite films in an air environment by the blade-coating method¹⁷. Therefore, the certified PCE of a mini-module with an aperture area of 63.7 cm² is 16.4%¹⁷. Introducing a strong Lewis base (sLBA) into the perovskite precursor solution, especially the sLBA additives with high

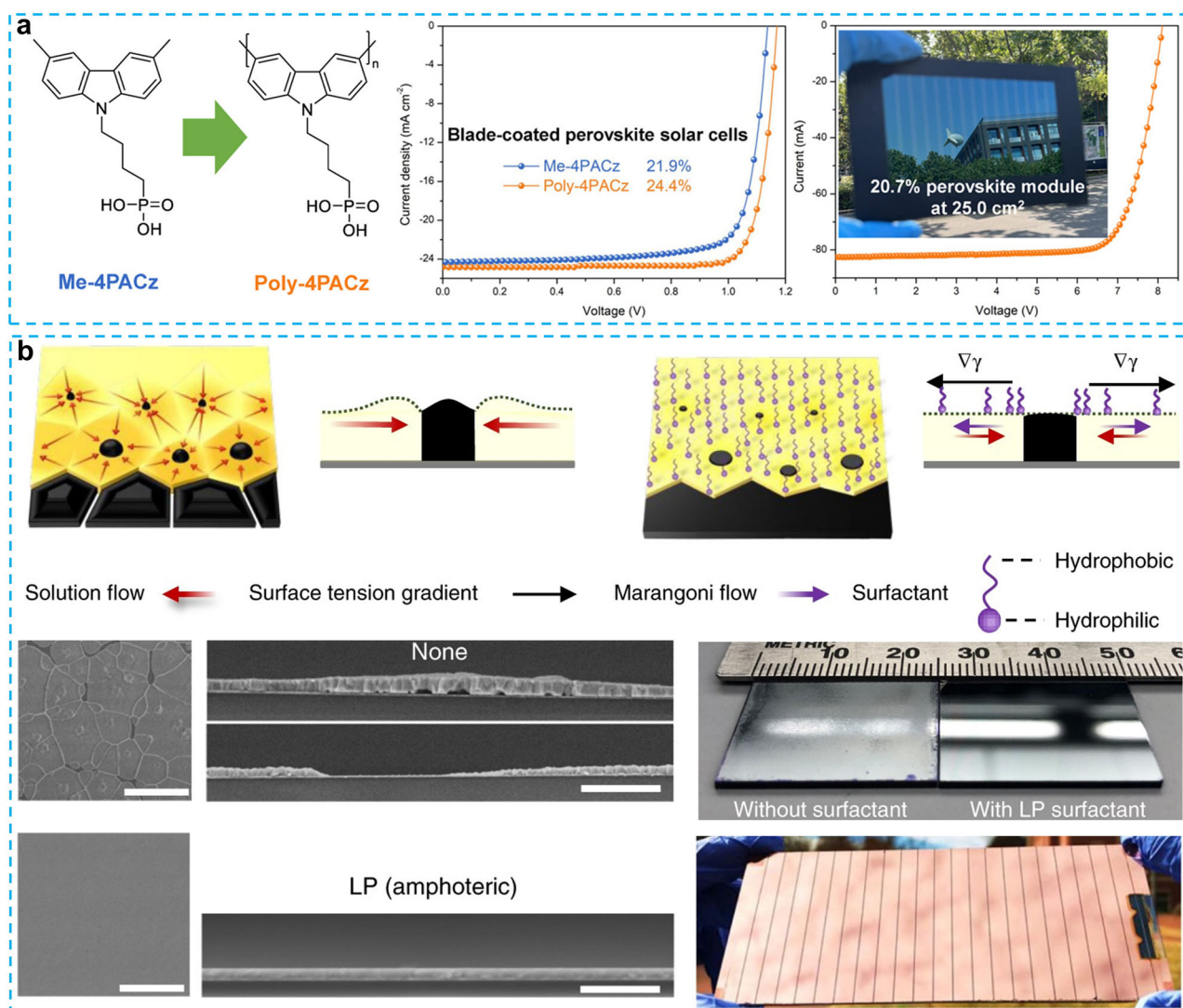


Fig. 5 | Fast nucleation by interface energy engineering. **a** Poly-4PACz was designed and synthesized to replace the Me-4PACz hole transport layer to improve wettability and increase interface energy. Reprinted with permission from ref. 36, copyright 2024 Elsevier. **b** The LP (zwitterionic surfactant) is used to increase the

solution's wettability and interface energy, thereby increasing the number of crystal nuclei to improve the compactness and uniformity of the film. Reprinted with permission from ref. 35, copyright 2019 Springer Nature.

boiling point and strong coordination, can significantly enhance the stability of the solution^{15,38,39}. This is because the strong coordination between sLBA and PbI₂ can form PbI₂-sLBA intermediate phase, which can exist stably (such as DMSO, DPSO, NMP, and DMPU)^{15,38,39}. However, the highly volatile, weakly coordinated solvents usually do not form a new intermediate phase with PbI₂, so the solution is prone to aging and affects performance^{15,38,39}. For example, Wang et al. emphasized in the "Method" that the precursor solution (highly volatile, weakly coordinated solvent) should not exceed 48 h⁹². To avoid this situation, a small amount of sLBA is usually added to this highly volatile, weakly coordinated solvent; for example, a small amount of DMSO or NMP is added to 2-ME to solve the instability problem of this solution^{36,74,93,94}. In addition, this highly volatile, weakly coordinated solvent has limited solubility for many additives, such as quaternary ammonium salts, making it difficult to be compatible with the additive engineering of lab-scale devices.

Interface energy engineering. According to the nucleation energy barrier Eq. (1) and nucleation rate Eq. (2), increasing the interface energy between the solution and the substrate can reduce the nucleation energy barrier and improve the nucleation rate and the number of nuclei. The interface energy is negatively correlated with the contact angle, thus

reducing the contact angle can increase the interface energy. Therefore, improving the wettability of the precursor solution on the bottom substrate can increase the interface energy, nucleation rate and number of nuclei, thereby improving the compactness and uniformity of the large-area film. There are two ways to improve wettability, one is to select a substrate with good wettability (substrate engineering), and the other is to introduce surfactants into the solution (surfactant engineering).

Substrate engineering. Substrate wettability is a critical factor in the fabrication of perovskite solar cells, as it significantly influences the interface energy and the quality of the resulting film. Ren et al. have synthesized a novel hole transport layer molecule, Poly-4PACz, derived from carbazole phosphonic acid (PACz)³⁶. This polymer enhances the wettability of the perovskite precursor solution during blade-coating, thereby increasing the interface energy with the substrate (Fig. 5a)³⁶. In addition, benzylhydrazine dihydrochloride (BHC), DMSO, and MAH₂PO₂ were introduced to address the problem of too fast growth rate caused by increased crystal nuclei³⁶. As a result, a compact and uniform film with reduced trap density was obtained, culminating in the creation of a 25 cm² (aperture area) mini-module with an impressive PCE of 20.7%³⁶. In comparison to organic layers such as PTAA and self-assembled molecules (SAMs), inorganic oxides like

SnO_2 , TiO_2 , and NiO_x offer superior wettability and interface energy for the precursor solution. This is attributed to the ability of the -OH groups in inorganic oxides to form strong hydrogen bonds with the N and H atoms in the hybrid perovskite, surpassing the van der Waals interactions in organic layers^{55–57,95–97}. For instance, Chang et al. successfully fabricated a mini-module on a SnO_2 substrate (delay crystallization by methylammonium chloride (MACl)) with a PCE of 17.54% and an active area of 10.93 cm^2 using a blade-coating method³⁷. Despite these advancements, the quest for an organic transport layer that can accommodate PSMs remains challenging. The requirements for such a layer are multifaceted: it must possess good solubility in organic or water solvents, exhibit high stability and carrier mobility, maintain high transmittance, and ensure excellent wettability with the perovskite precursor, all while matching the energy levels appropriately. Currently, the fabrication of PSMs, particularly those exceeding 200 cm^2 , predominantly relies on inorganic transport layer substrates. Achieving large-area films with the desired compactness and uniformity on SnO_2 and TiO_2 substrates is complex, and while these materials are integral to the formal structure of PSMs, they often fall short in stability^{1,3–5}. Although NiO_x films can be produced with the desired characteristics through magnetron sputtering or electron beam evaporation, their high surface reactivity poses a threat to the long-term stability of PSMs⁷⁰. In summary, the search for the perfect substrate that can deliver both high performance and long-term stability in large-area PSMs is ongoing, with current solutions presenting a trade-off between these two critical attributes.

Surfactant engineering. Surfactants, molecules endowed with both hydrophilic and hydrophobic domains, are adept at self-assembly on solution surfaces, projecting their hydrophobic groups outward. This configuration substantially diminishes the surface tension at liquid–solid interfaces, enhancing the liquid's spread on solid substrates^{98–100}. Notably, surfactants have long been utilized in the printing industry to polish the quality of printed films^{101,102}, a practice now adapted for the fabrication of PSMs^{35,59,101–106}. Previously, Deng et al. incorporated a trace of the zwitterionic surfactant 1- α -Phosphatidylcholine (LP) into the perovskite precursor solution⁵³. The LP notably improved the solution's infiltration on hydrophobic PTAA substrates by modulating the flow dynamics, which in turn escalated the nucleation rate and the number of nuclei, culminating in the enhancement of uniformity and density in large-area films (Fig. 5b)³⁵. Their method yielded a mini-module with an aperture area of 57.2 cm^2 and a PCE of 14.6% using the blade-coating technique³⁵. Following this research, Ren et al. elucidated the influence of amphoteric surfactants with varying alkyl chain lengths on the blade-coating preparation of perovskite films⁵⁹. They discovered that longer alkyl chains are instrumental in lowering the critical micelle concentration, allowing minimal surfactant additives to significantly benefit the rheological, hydrodynamic, and viscoelastic properties of the coating ink⁵⁹. This advancement led to improved wettability and, consequently, more uniform and compact films. Employing the blade-coating method, they fabricated a mini-module with carbon electrodes, an active area of 50 cm^2 , and a PCE of 17.05%⁵⁹. Later, Wang et al. conducted a comprehensive investigation into the effects of integrating quaternary ammonium cations (QACs) into perovskite ink for the preparation of large-area perovskite films via blade-coating¹⁰¹. Their findings indicated that double-chain QACs significantly favored the regulation of crystallization growth and defect passivation in perovskite, resulting in a mini-module with an extensive active area of 175 cm^2 and an impressive PCE of 19.74%¹⁰¹. While surfactants have proven advantageous in improving solution wettability and interface energy, their application is currently limited to specific perovskite systems. The quest for universal surfactants remains, as their development is key to advancing the large-scale production of PSMs. This pursuit underscores the need for surfactants that can universally enhance the fabrication process, thereby propelling the field toward the realization of high-efficiency, large-area PSMs.

Seed engineering. In addition to the above methods, seed engineering is the most widely used strategy in accelerating nucleation. Seed

engineering can be divided into two categories according to the difference in preparation methods. One is to add seed to the precursor solution directly. The other is depositing a seed layer on the substrate before coating perovskites. In the first case, the seed powder was first synthesized and then added to the precursor solution. In this way, the chemical properties of the precursor solution will be changed. The colloidal size in the solution will be increased, so the gap between the colloidal size in the solution and the critical crystal nucleus size will be narrowed. According to Eqs. (1) and (2), this will lead to a decrease in the energy barrier of crystal nucleation, an increase in the nucleation rate, and an increase in the number of crystal nuclei. Furthermore, the seed usually could change the homogeneous nucleation into heterogeneous nucleation in the solution. According to Eqs. (1) and (2), this will also lead to a decrease in the nucleation energy barrier and increase the nucleation rate and the number of nuclei. Therefore, this method can significantly accelerate crystal nucleation. In addition, the large-area perovskite films prepared in this way usually have preferred crystal orientation, more uniform component distribution, lower trap density, higher carrier mobility, and longer carrier diffusion length. For example, Rana et al. reported that introducing KPB_2Br_5 seeds into the precursor solution could accelerate nucleation and improve the uniformity of the film (Fig. 6a)²³. Moreover, the large-area films prepared using this method have better orientation and lower trap density. Thanks to these advantages, a mini-module with an active area of 57.5 cm^2 and a PCE of 16.22% was prepared based on the slot-die coating method²³.

In the second case, the seeds were first synthesized and deposited on the substrate before blade/slot-die coating perovskite. This method can improve the interaction between perovskite and the substrate, thereby reducing the contact angle and increasing the wettability. The decrease in contact angle will result in increased interface energy and thus reduced nucleation energy barrier, which increase the nucleation rate and nuclei number. Moreover, this seed layer can be used as a template to increase heterogeneous nucleation, thereby increasing the nucleation rate and number of nuclei. In addition, this method can effectively prevent the formation of pinholes at bottom interface and reduce the defect density at the buried interface. For example, Li et al. reported that pre-coating a layer of trimethylsulfide iodine ($(\text{CH}_3)_3\text{SI}$) on a NiO_x substrate not only could serve as a seed template to accelerate nucleation and increase the number of nuclei but also effectively reduce the bottom interface defects (Fig. 6b)¹⁰⁷. Finally, they fabricated a mini-module with a designed area of 100 cm^2 and a PCE of 20.03%¹⁰⁷.

Seed engineering can significantly improve the nucleation rate and increase the number of nuclei, thereby improving the uniformity and crystal quality of large-area perovskite films. However, these seeds remain in the perovskite film, which may lead to uneven distribution of components. In addition, these one-dimensional crystals stay on the perovskite film. They may serve as seed templates for phase transformation into one-dimensional perovskite (δ phase) during long-term operation, thereby inducing phase transition (from α phase to δ phase) of perovskite and deteriorating the stability of PSMs.

To fabricate PSMs, a single approach to nucleation often proves insufficient. A multifaceted strategy is essential to significantly optimize nucleation and augment the number of nuclei. For instance, integrating GASP, solvent engineering, and surfactant engineering can collectively enhance the process. GASP and solvent engineering synergistically work to hasten solvent evaporation, swiftly leading to solution supersaturation and an increased nucleation rate. Meanwhile, surfactant engineering optimizes wettability and interface energy, lowers the energy barriers for nucleation, and thus further accelerates the process. Multi-strategy has gained considerable attention over the past two years, particularly in the production of PSMs that boast high PCE. Such a holistic approach, as detailed in Table 1, leverages the strengths of each individual strategy to overcome the limitations of traditional methods, thereby setting a new standard for the blade/slot-die coating technique in PSM development.

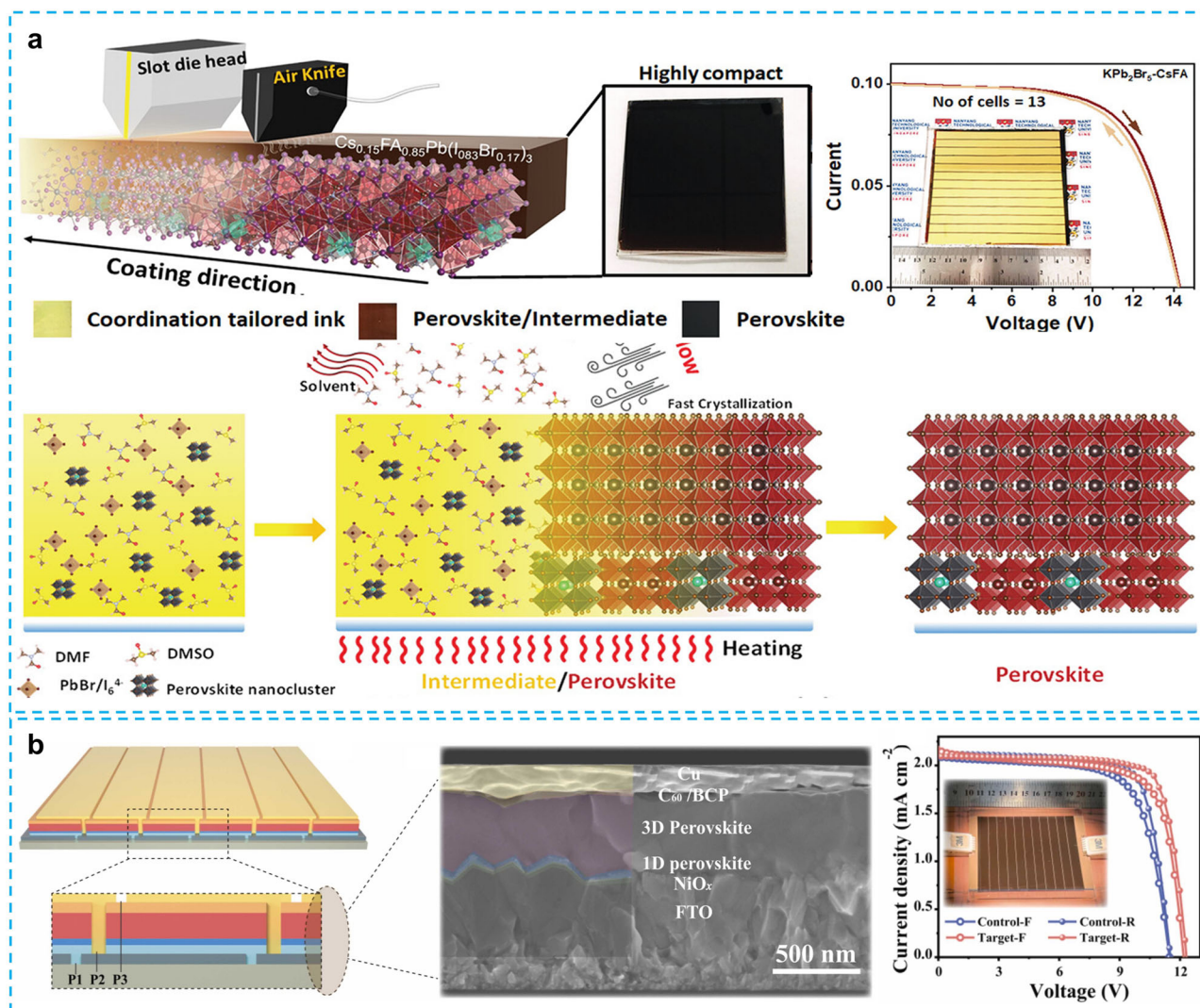


Fig. 6 | Fast nucleation by seed engineering. **a** The KPb_2Br_5 seeds were introduced into the precursor solution to increase the nucleation rate and the number of nuclei, improving the uniformity and crystal quality of large-area perovskite films. Reprinted with permission from ref. 23, copyright 2022 Wiley-VCH GmbH. **b** A seed

layer of $(\text{CH}_3)_3\text{SI}$ is pre-coated on the NiO_x substrate to accelerate nucleation and increase the number of nuclei, thereby improving the uniformity and crystal quality of large-area perovskite films. Reprinted with permission from ref. 107, copyright 2024 Elsevier.

Slow crystallization

After analyzing how to achieve accelerated nucleation according to the three key factors (temperature, concentration, and interface energy) of nucleation, we next analyzed how to delay crystal growth to improve crystal quality. According to the crystal growth rate Eq. (3), slowing crystal growth means reducing the solute precipitation rate. The additive engineering is the most effective way to reduce the solute precipitation rate. The introduced additives usually interact strongly with PbI_2 (Fig. 7a), and can even form a new intermediate phase. The new intermediate phase will lead to competitive crystallization, delaying crystal growth. According to the different ways of action of additives, they are mainly divided into strong Lewis base (sLBA) and non-sLBA.

sLBA additives. The most commonly used additive to delay crystallization is the introduction of sLBAs (difficultly volatile, strongly coordinated, and the content in the precursor solution is usually 10–30%)^{15,38,39}. In the Lewis acid–base theory, Lewis acid refers to molecules or ions that can act as electron pair acceptors, such as Pb^{2+} ions in perovskites^{15,38,39}. Lewis base additives (LBA) refer to molecules or ions that can act as electron pair donors, such as additive molecules containing N, O, S, and P elements^{15,38,39}. The coordination ability of LBA with Pb^{2+}

to form adducts can be judged by donor number (DN)^{15,38,39}. It is generally believed that the coordination between LBA and PbI_2 is greater than 18 kcal/mol, which is a high DN value, and less than 18 kcal/mol is a low DN value^{15,38,39}. The strong coordination of high DN value can form a stable PbI_2 -LBA intermediate phase, effectively reducing the solute precipitation rate and inhibiting the formation of the δ phase. The coordination effect of a low DN value is weak, which is conducive to the continuous growth of the crystal and makes it easy to form a larger grain size, which is usually used for the synthesis of a single crystal. The DN values of standard LBAs and their coordination with Pb^{2+} are shown Fig. 7b–d^{15,38,39,108,109}.

sLBAs of O-donors are the most widely studied^{47,91,93,110–114}. The most commonly used sLBA of O-donors is DMF, but the DN value (26.6) of DMF is not large, and the coordination with PbI_2 is not strong^{47,91,93,110–114}. It is easy to form one-dimensional intermediate phases such as $\text{MA}_2\text{Pb}_3\text{I}_8$ in the precursor, resulting in rod-like and needle-like films, thus affecting the density and uniformity of the films^{73,110}. To solve this problem, some other O-donors sLBA with larger DN values were added into DMF solvents, the most typical of which is DMSO^{73,110}. DMSO has a larger DN value (29.8) than DMF, forming stronger coordination with PbI_2 , resulting in PbI_2 -DMSO intermediate phase^{73,110}. MAI is inserted into the

Table 1 | High-performance PSMs were prepared by regulating the crystallization process in the blade/slot-die coating methods

Material	Area (cm ²)	PCE (%)	Method	Fast nucleation	Slow growth	J _{sc} (mA/cm ²)	V _{oc} (V)	FF(%)	Structure	Refs.
FA _{0.9} MA _{0.05} CS _{0.05} Pb(I _x Cl _{1-x}) ₃	27.22	22.0	Blade-coating	GASP + SSE	DMSO+MACI+CsCl + [Bemim]Cl	3.02	9.35	80.53	n-i-p	18
FA _{0.88} CS _{0.12} PbI ₃	205	15.30	Blade-coating	GASP + SSE	NMP+MACI	1.01	20.895	72	n-i-p	25
MA _{0.6} FA _{0.4} PbI ₃	50.0	19.20	Blade-coating	GASP + SE	DMSO + CBH + MAH ₂ PO ₂	1.53	16.1	79.8	p-i-n	21
FA _{0.9} CS _{0.1} PbI ₃	26.9	21.10	Blade-coating	SSE + SF + SE + SDE	DMSO + CBH+ZnCl ₂ + FACI+FAH ₂ PO ₃	2.83	9.38	79.5	p-i-n	24
MA _{0.7} FA _{0.3} PbI ₃	25	20.70	Blade-coating	GASP + SSE + SF + SE	DMSO + MAH ₂ PO ₂	3.3	8.11	77.3	p-i-n	36
FA _{0.85} CS _{0.17} PbI ₃	65	19.54	Slot-die coating	GASP + SSE	NMP+PbCl ₂	1.67	15.35	76.1	n-i-p	19
FA _{0.87} CS _{0.08} Pb _{1.02} I ₃	174	18.60	Slot-die coating	VASP + SSE + SF	NMP+MACI+BMIMBF ₄	1.14	20.74	78.40	p-i-n	141
FA _{0.91} CS _{0.09} PbI ₃	175	19.74	Slot die	GASP + SSE	NMP+MACI	0.81	30.92	79.14	p-i-n	101
FAMAPbI _{3-x} Cl _x	171.5	18.73	Slot die	GASP	NMP+MACI+BMIMBF ₄	0.77	30.67	79.01	p-i-n	142
FAMAPbI _{3-x} Cl _x	100	20.03	Blade-coating	VASP + SSE + SE + SDE	NMP+MACI	2.11	12.31	77.12	p-i-n	107
FA _{0.95} MA _{0.05} PbI _{2.85} Br _{0.15}	81	20.15	Blade-coating	GASP + SSE + SE	DMI+MACI	1.71	15.46	76.3	n-i-p	94
FA _{0.97} CS _{0.03} PbI ₃	100	17.7	Blade-coating	VASP + SSE	NMP+MACI	1.56	15.58	72.78	n-i-p	143
FAPbI ₃	15.64	21.9	Blade-coating	GASP + SSE + SE	DMSO+MACI	3.14	9.00	77.55	n-i-p	144
FA _{0.35} MA _{0.65} PbI ₃	25	19.5	Blade-coating	GASP + SF + SE	DMSO + MAH ₂ PO ₂	3.13	8.14	76.7	p-i-n	145
FAPbI ₃	36	19.26	Blade-coating	VASP + SE	DMSO+MACI	2.99	8.55	75.05	n-i-p	146
FA _{0.85} CS _{0.15} PbI _{2.48} Br _{0.54}	58.5	19.27	Slot die	GASP + TE	DMSO	1.86	13.7	75	n-i-p	147
FA _{0.8} MA _{0.15} CS _{0.05} PbI ₃	25	21.6	Blade-coating	TE + SSE	DMSO+BPYBF ₄	3.83	7.184	78.5	n-i-p	148
FAPbI ₃	36.4	20.33	Blade-coating	GASP + SSE + SE	MACI+ACCI	2.04	11.79	79.89	n-i-p	149
Perovskite-perovskite	20.25	24.5	Blade-coating	GASP	DMSO + AAH	1.86	17.26	77.5	Tandem	22
Perovskite-perovskite	20.25	23.3	Blade-coating	GASP	DMSO + FSA	1.79	16.97	76.5	Tandem	26
Perovskite-perovskite	14.3	21.6	Blade-coating	GASP + SDE + SF	DMSO + BHC+MAPbCl ₃	3.5	8.06	76.6	Tandem	27

TE temperature engineering, SSE substrate engineering, SF surfactant engineering, SE seed engineering, SDE seed engineering, CBH carbohydrazide, FACI formamide chloride, FAH₂PO₃ formamide hypophosphite, DSPC 1, 2-distearoyl-sn-glycero-3-phosphocholine, MAH₂PO₂ methylammonium hypophosphite, BMIMBF₄ 1-butyl-3-methylimidazolium tetrafluoroborate, DMI 1,3-dimethyl-2-imidazolidinone, BPYBF₄ 1-butylpyridine tetrafluoroborate, ACCI quaternary ammonium chloride, [Bemim]Cl 1,3-bis(cyanomethyl)imidazolium chloride, AAH aminoacetamide hydrochloride, FSA formamide sulfonic acid, BHC benzyhydrazine hydrochloride.

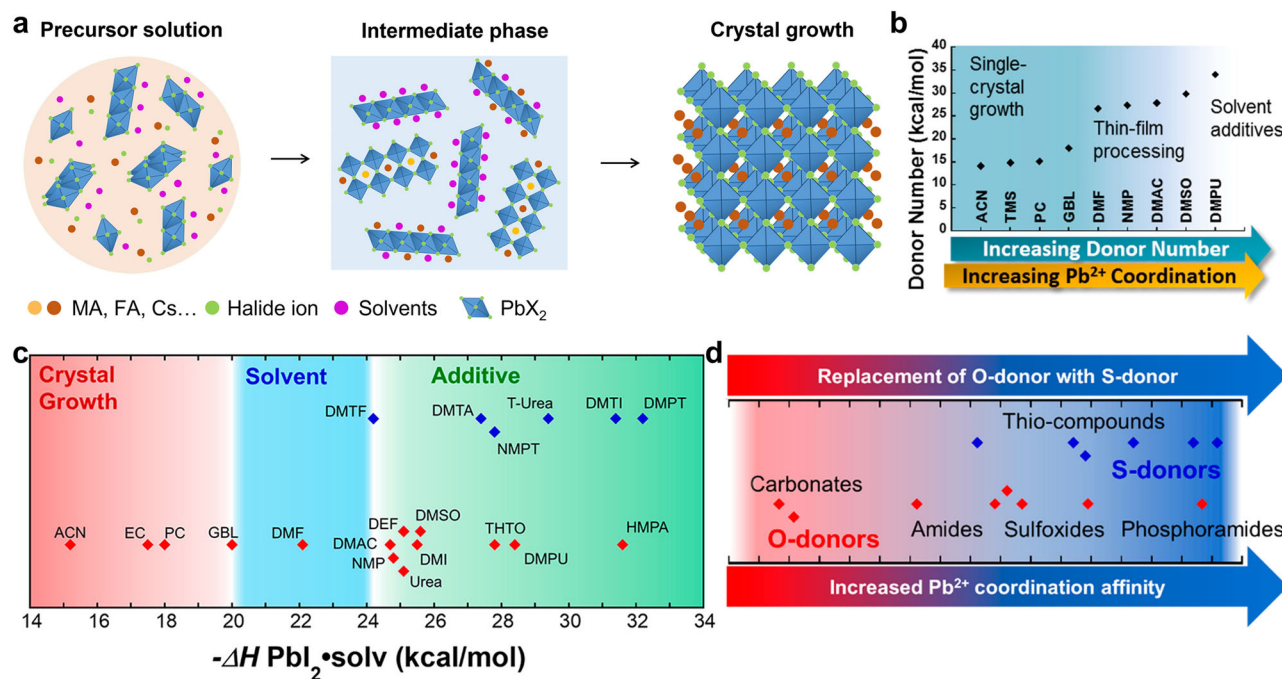


Fig. 7 | Slow crystal growth by strong LBA additive. **a** The schematic diagram of the formation process of perovskite film prepared by solution method, including the precursor solution process, the intermediate phase process and the crystal growth process. Reprinted with permission from ref. 39, copyright 2022 Elsevier. **b** The relationship between the DN value of Lewis base and the coordination strength of

Pb²⁺. Reprinted with permission from ref. 108, copyright 2017 American Chemical Society. **c**, **d** are the calculated enthalpy changes and their additions to binding different Lewis bases with PbI₂. Reprinted with permission from ref. 109, copyright 2020 American Chemical Society.

layered PbI₂-DMSO intermediate phase to form the MAI-PbI₂-DMSO intermediate phase^{73,110}. Then the MAI-PbI₂-DMSO intermediate phase undergoes structural rearrangement and bond-breaking reaction under heating conditions to form MAPbI₃ and DMSO, in which DMSO will be removed by antisolvent and long-term annealing process^{73,110}. This MAI-PbI₂-DMSO intermediate phase has been confirmed by Fourier transform infrared spectroscopy and X-ray diffraction^{73,110}. The PbI₂-DMSO and MAI-PbI₂-DMSO intermediate phases can reduce the solute precipitation rate, which is beneficial for delaying crystal growth and improving crystal quality^{73,110}. In addition, DMSO is introduced into DMF or GBL solvent to form a mixed solvent system, which delays the solvent evaporation rate, narrows the crystal nucleation window, increases the number of crystal nuclei, and reduces the film's pinholes, thereby improving the film's uniformity^{73,110}.

Some other sLBAs with stronger O-donors have also been introduced into the DMF solvent system^{19,91,115–120}. For example, Bu et al. introduced NMP into perovskite ink with DMF as a co-solvent, and the reaction of NMP with PbI₂ can form PbI₂-NMP intermediate phase, which can directly form α -FAPbI₃ phase by reacting with FAI, avoiding the formation of non-photoactive δ -FAPbI₃ phase (Fig. 8a)¹⁹. This is mainly because the energy barrier from the PbI₂-NMP intermediate phase to the α -FAPbI₃ phase is 0.13 eV, which is less than the energy barrier from the δ -FAPbI₃ phase to the α -FAPbI₃ phase of 0.64 eV¹⁹. The decrease of the phase transition energy barrier accelerates the formation of the α phase, thereby accelerating nucleation, increasing the number of nuclei, and improving the smoothness and compactness of the film¹⁹. Then, they proved that the use of DMF alone makes it easy to form a one-dimensional intermediate phase of FA₂Pb₃I₈-4DMF, which leads to a large number of one-dimensional needle-like crystals in the subsequent annealing process, thus affecting the uniformity and compactness of the film¹⁹. Incorporating NMP to form the PbI₂-NMP intermediate phase has been instrumental in generating numerous spherical nuclei, notably diminishing the prevalence of one-dimensional, needle-like crystals¹⁹. This strategic approach enhances the film's compactness and uniformity,

as evidenced by the successful fabrication of a mini-module with an active area of 65 cm² and a PCE of 19.54% through slot-die coating¹⁹. Further advancements by Yang et al. involved the integration of DPSO into the perovskite ink (Fig. 8b), which, as corroborated by density functional theory calculations and FTIR, exhibits stronger coordination with PbI₂ compared to DMF and NMP⁷⁴. This robust interaction fosters a stable intermediate phase and mitigates the solute precipitation rate, thereby decelerating crystal growth⁷⁴. The culmination of this research was the production of a mini-module with an active area of 20.77 cm² and a certified quasi-stabilized PCE of 16.63%⁷⁴.

It is crucial to note that substituting secondary sLBAs for DMF should be carefully controlled, as a complete replacement hinders the transition from the intermediate to the perovskite phase, adversely affecting film uniformity and density. The high boiling point of sLBAs necessitates elevated temperatures for residual removal, which can degrade crystal quality and risk perovskite decomposition. In essence, sLBAs with significant DN establish potent interactions with PbI₂, curtailing the solute precipitation rate and prolonging crystallization. Currently, they prevent the presence of non-perovskite phase byproducts, such as the δ phase, and contribute to a dense nucleation, a narrow grain distribution, a smooth film surface, and an overall improvement in the compactness and uniformity of the large-area perovskite film. Nevertheless, incomplete conversion of the sLBA-PbI₂ intermediate phase can induce phase heterogeneity in large-area films. Moreover, residual sLBA at the film's bottom interface post-annealing can escalate defect formation and pinholes, undermining the long-term stability of PSMs.

Non-sLBA additives. The most commonly used non-sLBA additives are volatile halide additives. Especially for volatile chlorides, Cl⁻ ions can not only effectively inhibit the formation of the solvation phase, but also form an intermediate phase containing Cl⁻, which leads to competitive crystallization and reduces the solute precipitation rate^{37,121–137}. For example, Chang et al. found in the two-step blade-coating preparation of large-area PSMs³⁷, the introduction of MAcl in the second step can promote

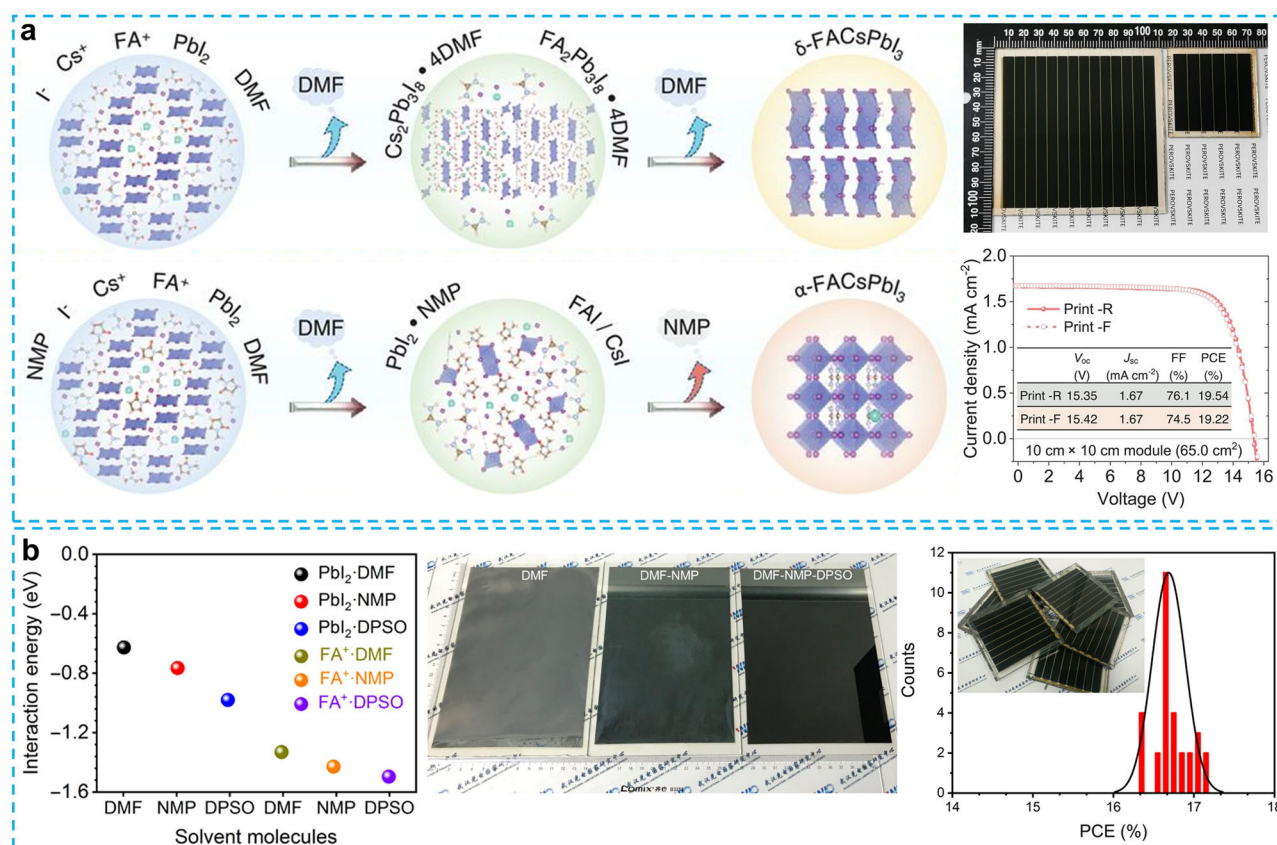


Fig. 8 | Slow crystallization by sLBA additives. **a** NMP was introduced to prepare FACsPbI₃ in a pure DMF system. The hydrogen bonding between NMP and FAI can promote the formation of the α-FACsPbI₃ phase. The strong coordination between NMP and PbI₂ can delay the crystallization to improve the crystal quality of the film. Reprinted with permission from ref. 19, copyright 2021 American Association for

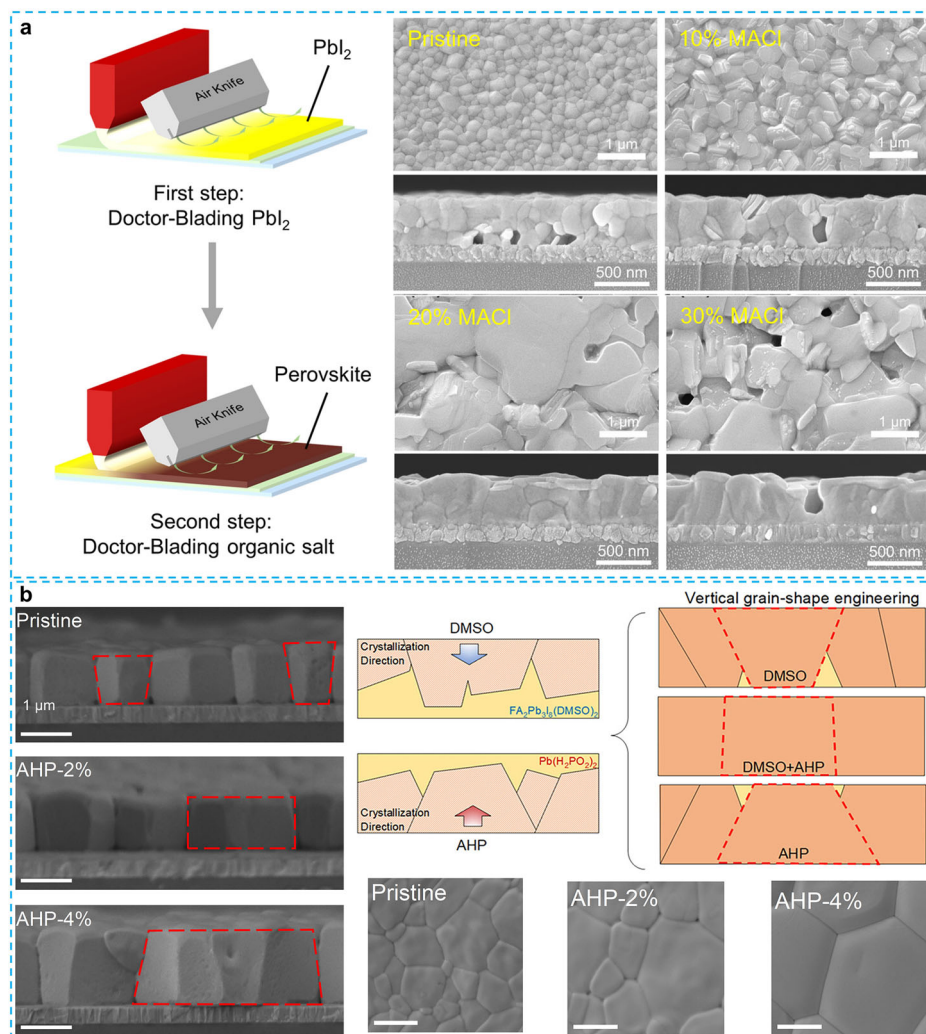
the Advancement of Science. **b** DMSO was introduced to prepare FACsPbI₃ in the DMF/NMP system. Compared with DMF and NMP, DMSO has stronger coordination with PbI₂ to delay crystallization, which is more beneficial to improving the crystal quality of the film. Reprinted with permission from ref. 74, copyright 2021 American Association for the Advancement of Science.

the formation of the α-FAPbI₃ phase and form the MAPbCl₃ intermediate phase. The MAPbCl₃ intermediate phase led to competitive crystallization (Fig. 9a), thus delaying crystal growth³⁷. Finally, a mini-module with an active area of 10.93 cm² and a PCE of 17.54% was prepared³⁷. In addition, some non-volatile and non-LBA additives can also interact with AX or BX₃, reducing the solute precipitation rate to delay crystallization^{7,10,138–140}. For example, Shi et al. reported the introduction of anilinium hypophosphite (AHP) additive in the precursor solution (Fig. 9b)¹⁴⁰. The H₂PO₂[−] in AHP reacts with Pb²⁺ to form a Pb(H₂PO₂)₂ intermediate phase, which not only leads to competitive crystallization but also reduces the solute precipitation rate. Therefore, the intermediate phase of Pb(H₂PO₂)₂ can delay crystallization, increasing grain size and improving crystal quality. In addition, the intermediate phase of Pb(H₂PO₂)₂ promoted vertical growth of grains to enhance orientation. Finally, based on the blade-coating method, they fabricated a mini-module with an aperture area of 14 cm² and a PCE of 18.06%¹⁴⁰. However, non-sLBA additives are often only suitable for specific processes and specific perovskite systems, and their versatility is poor. In addition, most of the currently reported non-sLBA additives are small molecules. The volatile non-sLBA additives can leave the film entirely after annealing, which will not negatively impact the film's stability. However, it is necessary to precisely control its concentration to avoid pinholes in large-area perovskite films during the annealing process. In particular, for volatile non-sLBA additives such as MACl, when the MACl content is less than 10%, the MACl leaves the film entirely after annealing¹³³, which will not affect the film's stability and band gap. However, when the content of MACl exceeds 20%, MA⁺ and Cl[−] of MACl will enter the lattice after annealing, causing FAPbI₃ lattice shrinkage and

band gap increase^{118,128,133}. Moreover, under high concentration the rapid volatilization of MACl might lead to pinholes in the final film. To solve this problem, Park et al. proposed combining long-chain alkylammonium chlorides (RACl) and MACl to delay the volatilization rate and reduce pinholes⁹. In addition, a high concentration of MACl will lead to Cl[−] aggregation, resulting in poor film uniformity. To mitigate this issue, Ding et al. proposed the introduction of 1,3-bis (cyanomethyl) imidazolium chloride ([Bcmim]Cl)-doped MACl, which can effectively avoid Cl[−] aggregation, thereby significantly improving the uniformity of large-area films¹⁸. Furthermore, non-sLBA additives such as MACl are extremely sensitive to the atmosphere, and the optimal concentration of the same perovskite component is also different under different temperature and humidity. For non-volatile non-sLBA additives, although they can stay at the grain boundary/interface and passivate defects after the reaction, they are prone to aggregation or shedding under long-term light and high temperature due to their poor stability, thus damaging the long-term stability of PSMs.

In preparing large-area PSMs by the blade/slot-die coating method, a wide processing window is often required, that is, slow crystallization. While the delay crystallization by only using strong LBA is limited. Therefore, some non-LBA additives based on strong LBA are added to further delay crystallization. For example, Fei et al. developed a multi-additive combination strategy to delay crystallization and prepare high-efficiency PSM. Using two strong LBAs and three non-LBAs to delay crystallization, they prepared a mini-module with an aperture area of 26.9 cm² and a certified PCE of 21.8% by a blade-coating method²⁴. Other multi-additive combination strategies to delay crystallization for PSMs with high PCE are shown in Table 1^{18,19,21,22,24–27,36,94,101,107,141–149}.

Fig. 9 | Slow crystallization by non-sLBA additives. **a** The experimental schematic diagram of FAPbI₃ films prepared by the two-step method and the effect of different concentrations of MACl on the morphology. Reprinted with permission from ref. 37, copyright 2023 Elsevier. **b** The film morphology and mechanism diagram of AHP are introduced. Reprinted with permission from ref. 140, copyright 2023 Elsevier.



Outlook

Achieving large-area perovskite films with exceptional compactness, uniformity, and crystalline quality is paramount for enhancing the efficiency and longevity of PSMs. Studies investigating the crystallization dynamics of these films, particularly those produced via blade/slot-die coating methods, indicate that rapid nucleation, leading to a higher density of nuclei, significantly enhances film compactness and uniformity. Conversely, delaying crystal growth proves beneficial for refining crystal quality.

During nucleation, key factors include temperature, solute concentration, and interfacial energy, while growth is primarily influenced by temporal variations in the solution's supersaturation level. In response to these fundamental factors, various crystallization regulation strategies have been developed, such as GASP, VASP, and solvent engineering for promoting rapid nucleation, and sLBA and non-sLBA additives for facilitating slow crystallization. It is important to note that employing a single strategy to regulate crystallization kinetics has limitations, and a combination of multiple strategies is typically required to achieve optimal results in terms of both rapid nucleation and slow crystallization.

While remarkable progress has been made in this field, many challenges with regard to the manipulation of crystallization kinetics in large-area perovskite film fabrication remain to be addressed in achieving high-efficiency, long-lasting, and reproducible PSMs for practical application. To further advance the development of large-area PSMs, we give a perspective on future research directions as outlined below (Fig. 10).

Interface engineering

The interface engineering mentioned above plays a crucial role in the nucleation and growth of large-area perovskite films. Large-area PSMs commonly utilize substrates such as NiO_x and SnO₂. NiO_x is typically prepared via magnetron sputtering or electron beam evaporation¹⁵⁰, SnO₂ through chemical bath deposition^{12–14}. As previously discussed, inorganic oxide transport layers exhibit superior wettability, thereby enhancing the uniformity of large-area perovskite films. Consequently, efforts have been directed towards developing alternative inorganic transport layers, such as SnO₂, TiO₂, ZnO, Cu₂O, WO_x, and Zn₂SnO₄, utilizing methods like magnetron sputtering or vacuum deposition. However, the high surface reactivity of oxide transport layers like NiO_x poses challenges^{151,152}. Strategies such as depositing Al₂O₃ or SiO₂ layers via atomic layer deposition or surface modification with SAMs have been explored to mitigate surface reactivity^{151,152}, thereby enhancing the performance and stability of large-area PSMs.

Development of new processing technologies for SAMs in large-area PSMs

SAMs exhibit desirable traits such as high transmittance, energy level alignment, and high carrier mobility, making them valuable in high-efficiency lab-scale PSCs with p-i-n structures. Amphiphilic molecular SAMs, such as (2-(4-(bis(4-methoxyphenyl) amino)phenyl)-1-cyanovinyl) phosphonic acid⁶¹, have shown promise in enhancing solution wettability and improving the uniformity of large-area perovskite films. In addition, the

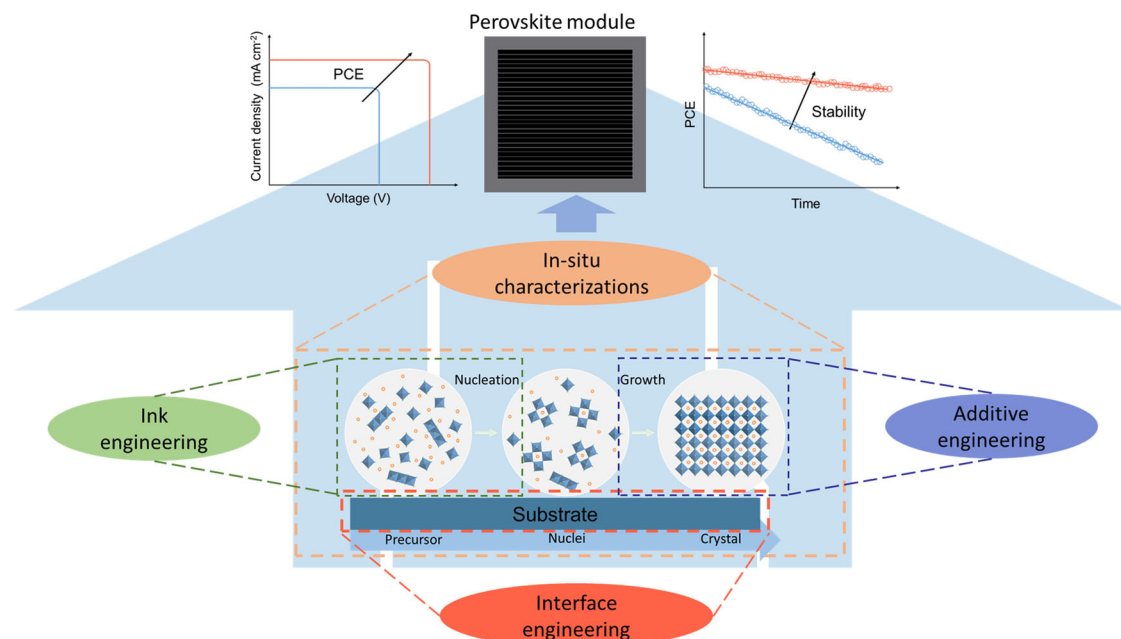


Fig. 10 | Future opportunities in manipulating crystallization kinetics towards high-efficiency and long-lifetime PSMs.

strong coordination between phosphonic acid and Pb^{2+} can effectively delay crystallization and enhance crystal quality. However, the spin-coating method commonly used for SAMs in lab-scale PSCs is not suitable for large-area PSMs. Hence, alternative methods like vacuum deposition, thermal evaporation, or dip-coating can be explored for large-area SAM preparation. In addition, the development of new SAMs suitable for large-area PSMs, such as polymerized SAMs (such as poly-4PACz)³⁶, can effectively improve film uniformity and crystal quality.

Development of single-crystal transport layer substrates

Novel substrates such as single-crystal SrSnO_3 and TiO_2 can be developed using techniques like magnetron sputtering or other scaled-up methods^{75,153}. The high lattice matching between these single-crystal substrates and perovskite facilitates epitaxial growth, transforming the traditional island and layered mixed growth model of perovskite films into a layered growth model. This transition significantly enhances crystal quality and uniformity while burying bottom interface defects and reducing residual sLBA at the bottom interface, thereby enhancing the long-term stability of PSMs.

Ink engineering

Developing single-crystal dissolution and recrystallization engineering

The process involves synthesizing high-purity perovskite single crystals initially, followed by their dissolution to prepare PSMs. The precursor solution produced through single-crystal dissolution and recrystallization engineering typically features larger colloidal particle sizes, facilitating accelerated nucleation and delayed growth. Furthermore, this engineering approach enhances the purity of the initial perovskite phase in the solution, thereby reducing the formation of solvation and intermediate phases and improving film uniformity and phase purity. In addition, single-crystal dissolution and recrystallization engineering effectively mitigate the adverse effects of raw material impurities on PSMs.

Additive engineering

Currently, sLBAs predominantly employed for growth delay consist of O- and S-donors. However, sLBAs featuring P-donors exhibit larger DN values, indicating stronger coordination with PbI_2 ^{38,39}. The exploration of zwitterionic molecules with enhanced interaction with perovskites¹⁵⁴, particularly those incorporating phosphate and amino functional groups akin to

phosphocholines¹⁵⁵, holds promise for further delaying growth and enhancing crystal quality. Moreover, the utilization of multi-component perovskites, known for their superior thermodynamic stability compared to single-component counterparts, presents opportunities for high-efficiency and long-lifetime PSMs. Nonetheless, multi-component perovskites often encounter challenges such as disparate crystallization rates and uneven component distribution. Therefore, it is imperative to find ideal additive molecules to regulate the crystallization rates among different components, thereby improving the uniformity of perovskite films. For instance, Liang et al. devised a novel 1-(benzenesulfonyl)pyrrole additive to modulate the crystallization rate among different A-site cations, resulting in the fabrication of homogeneous perovskite films⁷.

Despite the importance of additives, the additive residual left in the perovskite films after crystallization might cause issues such as uneven composition and deteriorated device stability. The reasons for additive residues are often diverse, but here we can primarily classify them into the following two main causes.

(1) In preparing large-area perovskite films by blade/slot-die coating, the sLBA additives such as DMSO are usually introduced to delay crystallization. The strong coordination between sLBA and PbI_2 leads to the formation of the sLBA- PbI_2 intermediate phase, which tends to remain in the resultant film after annealing, especially at the bottom interface of perovskite film (due to the volatilization of the top interface). However, the boiling point of DMSO is not high enough. During the long-term operation of PSMs, the DMSO- PbI_2 intermediate phase decomposes, and the volatilization of DMSO leads to local volume shrinkage and pinholes in the film, which will deteriorate the long-term stability of PSMs. The development of non-volatile sLBA could solve this problem. For example, carbonylhydrazide (CBH) will not volatilize during the whole annealing process and long-term operation process, and will always stay in the perovskite film, which will not lead to the formation of pinholes²¹. Another way to avoid pinholes is to completely remove DMSO by using VASP technology before annealing⁸⁷.

(2) In addition to sLBA additives, non-sLBA additives are usually introduced to further delay crystallization and passivate interface defects. However, the currently reported non-sLBA additives are usually small organic molecules or ionic liquids¹⁵⁶. The binding between small molecules and perovskite is not strong enough so that it is easy to desorb during the long-term operation under high temperature, resulting in new defects and reduced device stability. The development of macromolecules or

supramolecules to regulate crystallization and passivate interface defects can effectively solve this issue^{157,158}. For example, Zhang et al. reported a starch-polyiodide supermolecule as a bifunctional buffer layer at the perovskite interface, which can effectively inhibit ion migration and promote defect self-healing, thereby improving the long-term stability of PSCs¹⁵⁹.

In situ characterization

Many studies investigating the crystallization kinetics of perovskites through in situ characterization techniques have primarily focused on lab-scale PSCs utilizing the spin-coating method. However, the film preparation processes of spin-coating and blade/slot-die coating methods differ significantly, leading to observable distinctions in crystallization kinetics. Furthermore, variations in solvent or component composition further contribute to differences in crystallization kinetics among perovskite films. Hence, there is an urgent need to develop advanced in situ characterization techniques tailored for large-area PSMs prepared via blade/slot-die coating. Examples include in situ grazing-incidence wide-angle X-ray scattering, in situ photoluminescence spectroscopy, and in situ absorption spectroscopy. These techniques enable real-time monitoring of the intermediate processes involved in preparing large-area perovskite films via blade/slot-die coating, quantification of nucleation and growth rates, and identification of intermediate phases. Such in situ monitoring facilitates precise manipulation of crystallization kinetics, thereby enhancing the uniformity and crystal quality of large-area perovskite films. Moreover, it provides a novel avenue for improving the efficiency and stability of large-area PSMs.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Author contributions

Z.W., W.Z., Y.B., and H.W. conceived the idea. Z.W. prepared the first draft of the paper, Z.W., X.D., W.Y., and D.Q. prepared figures and tables. Z.W., X.D., J.Z., Y.C., L.H., H.W., H.C., G.Y., W.Z., Y.B., and H.W. revised the manuscript. All authors discussed the paper.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Wei Zhang, Yang Bai or Hui-Ming Cheng.

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